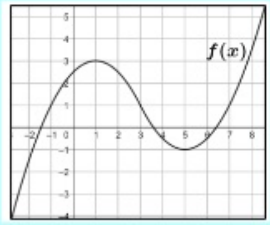
FUN-4.A.4 The graph of a function is concave up (down) on an open interval if the function's derivative is increasing (decreasing) on that interval.
FUN-4.A.5 The second derivative of a function provides information about the function and its graph, including intervals of upward or downward
concavity.
FUN-4.A.6 The second derivative of a function may be used to locate points of inflection for the graph of the original function.

5-6 Concavity and Points of Inflection

The first derivative of a function tells us many things about the original function. We know that $f'(x)$ can help us find where $f(x)$ has relative extrema as well as the intervals where $f(x)$ is increasing or decreasing.

What are some key intervals and important points on the graph of $f(x)$ that we should know?



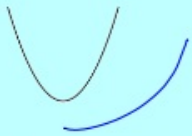
Important Features/Points of $f(x)$

1. Critical Points
2. Intervals where $f(x)$ is increasing/decreasing
3. Relative minimum/maximum
4. Intervals where $f(x)$ is concave up/down
5. Point(s) of inflection

What is concavity?

Concavity describes the curvature of a graph, indicating if it curves "up" (can hold water like a cup) or curves "down" (it can't hold water).

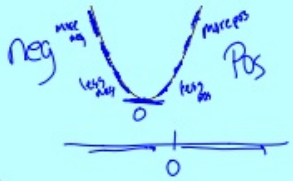
Concave up



Concave down

This parabola is concave up. What is $f'(x)$ doing?

slope of tan line



If f is concave up, then f' is **increasing**.

If f is concave down, then f' is **decreasing**.

This tells us:

$f''(x) > 0$ means f is **concave up**

$f''(x) < 0$ means f is **concave down**

What is a point of inflection?

$f(c)$ is a **point of inflection** for the graph of $f(x)$ if the graph of $f(x)$ is continuous at $x = c$ and the graph changes concavity (changes from concave up to concave down or concave down to concave up) at $x = c$.

So, the first derivative of a function can be used to identify critical points, increasing/decreasing intervals, and local minima/maxima, while the second derivative of a function can be used to identify intervals where the function is concave up/down and points of inflection.

So, let's summarize how we can find the important information for the graph of a function, $f(x)$:

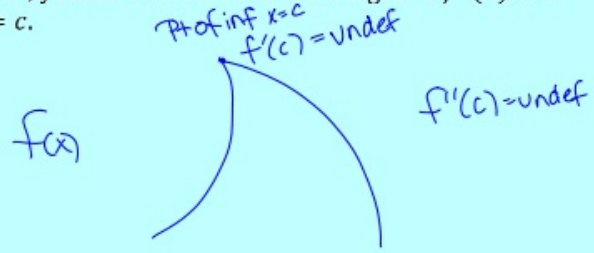
Given a continuous function $f(x)$...				
$f'(x) = 0$ or $f'(x)$ dne	$\longleftrightarrow$	$f(x)$ has a critical point		
$f'(x) \geq 0$	$\longleftrightarrow$	$f(x)$ is increasing		
$f'(x) \leq 0$	$\longleftrightarrow$	$f(x)$ is decreasing		
Relative (local) maximum	$\longleftrightarrow$	$f'(x)$ changes sign from $+$ to $-$		
Relative (local) minimum	$\longleftrightarrow$	$f'(x)$ changes sign from $-$ to $+$		
$f''(x) = 0$ or $f''(x)$ dne	$\longleftrightarrow$	**Possible** point of Inflection		
$f''(x) > 0$	$\longleftrightarrow$	$f'(x)$ is increasing	$\longleftrightarrow$	$f(x)$ is concave up
$f''(x) < 0$	$\longleftrightarrow$	$f'(x)$ is decreasing	$\longleftrightarrow$	$f(x)$ is concave down
Points of Inflection	$\longleftrightarrow$	$f''(x)$ changes signs $(+$ to $-$ or $-$ to $+)$ OR $f'(x)$ changes from increasing to decreasing or decreasing to increasing		

Two common **mistakes** students make when finding a point of inflection:

1.) Assuming that $f''(c) = 0$ means there is a point of inflection at $x = c$.

2.) Assuming that $f''(c)$ is undefined means there is not a point of inflection at $x = c$.

In both situations, you have to check that the signs of $f''(x)$ are changing at $x = c$.



To answer questions using $f'(x)$, we...	**To answer questions using $f''(x)$, we...**
1. Find any (possible) critical values for $f(x)$ by finding where $f'(x) = 0$ or is undefined.	1. Find any **Possible Points of Inflection** for $f(x)$ by finding where $f''(x) = 0$ or is undefined.
2. Use analysis (sign chart) to find intervals where $f'(x)$ is positive and negative.	2. Use analysis (sign chart) to find intervals where $f''(x)$ is positive and negative.
3. Interpret the sign chart to answer the question along with the proper explanation/reasoning.	3. Interpret the sign chart to answer the question along with the proper explanation/reasoning.

Notice that both involve the same process/concepts. It is important that you keep your work **well organized** and **well labeled**. It is also imperative that we know **WHICH** function, $f'(x)$ or $f''(x)$, we need for each problem.

Example 1: Find the open intervals where $f(x)$ is concave up and concave down and find any x-value(s) where $f(x)$ has a point of inflection. Give a reason for your answer.

$$f(x) = x^4 - 2x^3 - 12x^2$$

$f'(x) = 4x^3 - 6x^2 - 24x$

$f''(x) = 12x^2 - 12x - 24 = 0$

$12(x^2 - x - 2) = 0$

$12(x-2)(x+1) = 0$

$x = 2 \quad x = -1$

f'': $+$ | $-$ | $+$

f: -1 2

f is concave up on $(-\infty, -1)$ and $(2, +\infty)$ because $f''(x) > 0$ on these intervals

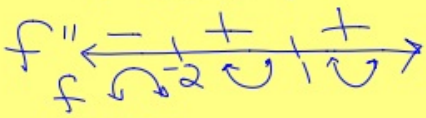
f is concave down on $(-1, 2)$ because $f''(x) < 0$ on this interval.

f has points of inflection at $x = -1$ because $f''(x)$ changes from pos to neg at this x-value and $x = 2$ because $f''(x)$ changes from neg to pos at this x-value.

Example 2: $f''(x) = (x+2)^3(x-1)^2$. Find any x-value(s) where $f(x)$ has a point of inflection.

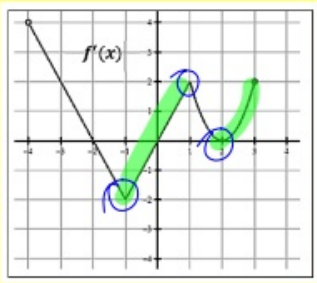
Possible points of inflection:

$x = -2 \quad x = 1$



f has a pt. of inf. at $x = -2$

Example 3: The graph of $f'(x)$ is at the right. Find the intervals where $f(x)$ is concave up/down and find any x-values where $f(x)$ is a point of inflection.



f conc up $\leftrightarrow$ f' inc.

f conc down $\leftrightarrow$ f' dec.

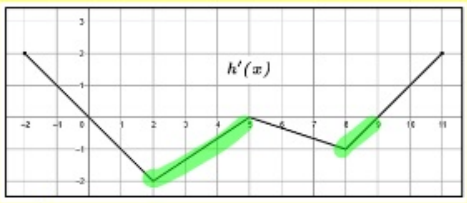
f conc. up on $(-1,1)$ and $(2,3)$

f conc. down on $(-4,-1)$ and $(1,2)$

f has a point of Inflection at $x=-1$, $x=1$, $x=2$

Example 4: The graph of $h'(x)$ is above. On what open interval(s) is $h(x)$ decreasing **and** concave up? Give a reason for your answer.

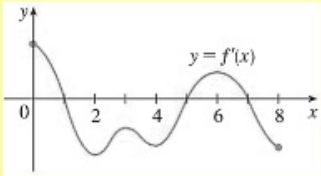
$h'(x) \leq 0$ h' inc.



$h(x)$ is dec. and concave up on $(2,5)$ and $(8,9)$ because $h'(x) \leq 0$ and $h'(x)$ inc. on these intervals.

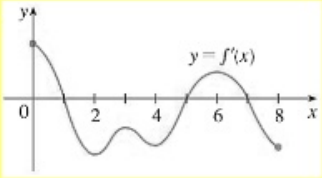
Example 5: The graph of $f'(x)$ is shown. Give a reason for each answer.

(a) On what intervals is $f(x)$ increasing?



(b) On what intervals is $f(x)$ decreasing?

(c) On what intervals is $f(x)$ concave up?



(d) On what intervals is $f(x)$ concave down?